

Jiayi (Elaine) Pan

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EDUCATION

M.S.E. in Industrial and Operations Engineering

Aug 2026 - Expected May 2027

University of Michigan, Ann Arbor, MI

- GPA: 4.0/4.0
- Relevant coursework: Dynamic Programming, Optimization and Computational Methods, Cost-Effectiveness Analysis in Health, Cognitive Ergonomics and Human System Integration, Discrete-Event Simulation

B.S.E. in Industrial and Operations Engineering

Aug 2023 - May 2026

University of Michigan, Ann Arbor, MI

- GPA: 3.8/4.0 | Summa Cum Laude

RESEARCH EXPERIENCE

Section of Digitally Enabled Care (SoDEC)

June 2026 - Present

Research Fellow, Department of Population Health, NYU Grossman School of Medicine, New York, NY

Advisors: Professor Angela Mastrianni, Professor Elizabeth Stevens, Professor Akhila Veerubhotla

- Engineered and iteratively refined prompts for an LLM-as-a-judge system that evaluates a patient-facing GLP-1 medication chatbot on clinical accuracy, safety, response quality, and patient journey progression
- Contributed to a governance framework for quality assurance of conversational patient-reported outcome collection; accepted for poster presentation (first author), 19th Annual D&I Conference in Health (2026)
- Developed a measurement framework for a digitally enabled primary care practice model, defining quality metrics and assessing AI-integrated diagnostic devices (e.g., digital otoscopes) on cost, feasibility, and implementation considerations
- Co-authoring a scoping review of physical activity and physical functioning as predictors of readmission after lung transplantation; presented at the 2026 Rusk Rehabilitation Summer Symposium

Center for Healthcare Engineering and Patient Safety (CHEPS)

Jan 2025 - May 2026

Research Assistant, University of Michigan, Ann Arbor, MI

Advisors: Professor Amy Cohn, William Pozehl

- Forecasted staffing demand across four state psychiatric hospitals, building regression models in Python on two years of census, absenteeism, and overtime data spanning multiple staff roles across day and night shifts
- Translated forecasts and clinical staffing rules into decision-support tools (interactive Plotly dashboards and Excel/VBA scheduling tools) used by hospital administrators for staffing planning and regulatory compliance
- Collaborated with 19 Michigan Medicine clinics to proactively identify open scheduling slots and coordinate with patients to advance appointment times, improving clinic capacity utilization and reducing patient wait times

Generative AI for Human Factors

Sept 2024 - Jan 2025

Research Assistant, University of Michigan, Ann Arbor, MI

Advisors: Professor Yili Liu, Maria Fields

- Conducted literature reviews examining the impacts of generative AI on user behavior, cognition, and trust to understand how emerging AI tools shape accessibility and human-technology interaction patterns in educational contexts
- Automated qualitative and text analysis workflows using MAXQDA and Python to systematically code large text corpora, enabling the efficient synthesis of user feedback, behavioral themes, and engagement patterns across diverse datasets
- Synthesized complex qualitative findings into clear, actionable research and design insights, presenting formal recommendations during weekly lab meetings to inform research priorities and human-centered technology development

INDUSTRY EXPERIENCE

Blue Cross Blue Shield of Michigan

May 2026 - Aug 2026

Product Architecture and Operations Intern, Detroit, MI

- Led the annual update of the HSA plan crosswalk used by multiple BCBSM departments to resolve billing issues, reconciling data across source systems and documenting the process for future cycles; completed in under three weeks
- Designed and developed a Tableau dashboard that automated manual monthly reporting of newly added structures, enabling Product Owners to access real-time data through a centralized self-service reporting solution
- Optimized multiple SQL queries supporting enterprise dashboards, reducing execution times by approximately 67% and improving reporting performance for recurring data refreshes and analysis
- Created end-to-end technical and business documentation, standardizing processes from Databricks SQL data sourcing through dashboard implementation to support Product Owners, developers, analysts, and knowledge transfer

C.S. Mott Children’s Hospital, Michigan Medicine

Aug 2025 - Dec 2025

Process Improvement Consultant, Ann Arbor, MI

- Evaluated neonatal intensive care unit (NICU) provider workload to identify cognitive overload risks and potential safety hazards by administering a NASA-TLX–based survey to capture critical factors not reflected in standard acuity metrics
- Conducted over 30 hours of structured clinical rounding and shadowing across multiple provider teams to map task flows, handoffs, and sources of cognitive load, contextualizing quantitative findings with real-world clinical observations
- Synthesized survey and observational findings into technical documentation and presented results to nursing leadership and clinical stakeholders, informing workload equity, workflow design, and more balanced patient assignment models to support hospital-wide patient safety initiatives

Henry Ford Health System (HFHS)

May 2025 - Aug 2025

Management Engineer Intern, Detroit, MI

- Mapped document approval workflows for the Henry Ford–Ascension joint venture rebrand and built a semi-automated triage process that reduces duplicate content for 100+ documents, saving 5 hours per review cycle
- Shadowed 40+ hours of patient calls in the ENT pod, identifying 30+ improvement opportunities and delivering recommendations that reduced average call handle time by 35% and improved scheduling accuracy by 20%
- Analyzed ENT provider schedule templates and appointment data in Excel to identify low-complexity visits that could shift to other providers or visit types, freeing physician capacity for higher-complexity cases

Business+Tech, Michigan Ross

May 2024 - Present

Tech Scholar, Ann Arbor, MI

- Designed and prompt-engineered “Jamie,” a retrieval-grounded AI coach that guides students through a five-stage persona and problem-statement framework, using a hidden rubric to gate progression and Socratic questioning
- Curated and co-led 4 Tech Forums on emerging topics including GenAI and FinTech, convening 300+ students with 30+ industry speakers and faculty judges for panel discussions and case competitions
- Increased participant engagement by 75% by redesigning the structure and problem framing of forums, hackathons, and Hack-the-Case competitions; 3 winning student solutions were adopted by partner organizations

SELECTED ACADEMIC & RESEARCH PROJECTS

Stochastic Scheduling of Neonatal Intensive Care Provider Teams

Winter 2026

IOE 543 (Scheduling), University of Michigan

Formulated a mixed-integer program in Gurobi to assign and sequence acuity-weighted NICU patients across provider teams, minimizing waiting, idle, and overtime costs. Extended it to a two-stage stochastic program via sample average approximation with lognormal service times, evaluated with 200-scenario Monte Carlo simulation and VSS/EVPI analysis. Developed local search heuristics (swap, relocate, reinsertion) that came within 2% of the exact optimum; results motivated a hybrid strategy pairing stochastic planning with heuristic real-time rebalancing.

Nissan Kicks HMI Predictive Modeling

Winter 2026

IOE 437 (Automotive Human Factors), University of Michigan

Developed and validated mathematical task-time models for a multi-level driver interface using KLM and SAE J2365 standards Engineered modular subroutines and predictive equations to analyze 10 distinct driver tasks, providing high-accuracy performance metrics to optimize menu-hierarchy efficiency and reduce the need for resource-intensive human-subject testing.

Managing Overcrowded Emergency Departments for Michigan Medicine’s Mott Children’s Hospital Based On Simulation Methods Performed in ProModel

Fall 2025

IOE 474 (Simulation), University of Michigan

Built and validated a ProModel discrete-event simulation of a pediatric emergency department to evaluate bed and provider utilization under varying staffing configurations, identifying a reduced-capacity policy that preserved throughput and wait-times.

PUBLICATIONS & PRESENTATIONS

Manuscripts in Preparation

- [1] **J. Pan**, A. Mastrianni, and E. Stevens, “Designing and validating an LLM-as-a-judge for quality assurance of a patient-facing GLP-1 medication chatbot,” in preparation.
- [2] B. Gu, **J. Pan**, and A. Veerubhotla, “Physical activity and physical functioning as predictors of readmission after lung transplantation: A scoping review,” in preparation.

Conference Presentations

- [1] **J. Pan**, A. Mastrianni, and E. Stevens, “Development and validation of an LLM-as-a-judge governance framework for quality assurance for conversational patient-reported outcome collection,” Poster (accepted), 19th Annual Conference on the Science of Dissemination and Implementation in Health, National Harbor, MD, Dec. 2026.

Other Posters

- [1] B. Gu, **J. Pan**, and A. Veerubhotla, “Scoping review of physical activity and physical functioning as predictors of readmission post-lung transplant,” Poster, Rusk Rehabilitation Summer Symposium, NYU Grossman School of Medicine, New York, NY, Aug. 2026.
- [2] **J. Pan**, D. Postiff, M. Retto, and J. Um, “NICU provider workload measurement and patient acuity scoring,” Poster, IOE Senior Design Expo, University of Michigan, Ann Arbor, MI, Dec. 2025.
- [3] **J. Pan**, A. Acharya, E. Nair, M. Stier, F. Zimmerman, W. Pozehl, and A. Cohn, “Database migration for state psychiatric hospital staffing and attendance tool,” Poster, CHEPS Symposium, University of Michigan, Ann Arbor, MI, Oct. 2025.

TEACHING EXPERIENCE

Introduction to Markov Processes (IOE 316)

Fall 2025 & Winter 2026

Undergraduate Instructional Aide, University of Michigan

- Provided in-class instructional support and office hours, assisting students with problem-solving and contributing to homework and exam development and grading for a 100-student course

SKILLS

- **Computing:** Python, R, SQL, VBA, MAXQDA, Power BI, Plotly, ProModel, Advanced Excel, MATLAB, C++
- **Methods:** Regression (Linear/Logistic), Markov Modeling, Discrete-Event Simulation, Six Sigma (Black Belt Trained)
- **Languages:** English (Native), Mandarin (Native), French (Fluent)

VOLUNTEER ACTIVITIES & COMMUNITY ENGAGEMENT

INFORMS @ UM

HFES @ UM

Alpha Pi Mu (IOE Honors Society)

Asian American Association

Chinese Student Association

Tech Plus Consulting

Business+Tech

Marketing Chair

Member

Member

Dance Lead

Member

Project Lead

Tech Scholar